Block Orthogonal Sparse Superposition Codes for Ultra-Reliable Low-Latency Communications

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***Abstract*— Low-rate and short-packet transmissions are important for ultra-reliable low-latency communications (URLLC). In this paper, we put forth a new family of sparse superposition codes for URLLC, called block orthogonal sparse superposition (BOSS) codes. We first present a code construction method for the efficient encoding of BOSS codes. The key idea is to construct codewords by the superposition of the orthogonal columns of a dictionary matrix with a sequential bit mapping strategy. We also propose an approximate maximum a posteriori probability (MAP) decoder with two stages. The approximate MAP decoder reduces the decoding latency significantly via a parallel decoding structure while maintaining a comparable decoding complexity to the successive cancellation list (SCL) decoder of polar codes. Furthermore, to gauge the code performance in the finite-blocklength regime, we derive an exact analytical expression for block-error rates (BLERs) of single-layered BOSS codes in terms of relevant code parameters. Lastly, we present a cyclic redundancy check aided-BOSS (CA-BOSS) code with simple list decoding to boost the code performance. Our experiments verify that CA-BOSS codes with the simple list decoder outperform CA-polar codes with SCL decoding in the low-rate and finite-blocklength regimes while achieving the finite-blocklength capacity upper bound within one dB of signal-to-noise ratio.**

***Index Terms*— Error correction codes, short packet transmissions, low-latency decoders.**

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I. INTRODUCTION

A. Motivation

THE aspiration for ultra-reliable low-latency communications (URLLC) is unrelenting for next generation wireless systems. URLLC is envisioned to ensure that a data packet is delivered within a very short time duration (e.g., within 1 ms) while satisfying very high reliability (e.g., the packet error rate is less than 10^{-6}) [2], [3], [4], [5], [6]. These stringent requirements of the latency and reliability are indispensable to support various mission-critical applications that demand prompt responses with ultra-high accuracy, including automated driving, industrial automation, and telesurgery [7], [8]. To deliver a data packet with extremely low-latency and high-reliability, it is essential to design a novel low-rate coded modulation technique that is fast decodable while achieving near-optimal performance in the finite-blocklength regime [9], [10]. The conventional approach for designing low-rate codes with short blocklenghs is to combine standard codes, such as turbo codes [11] and low-density parity-check (LDPC) codes [12], [13], with a simple repetition code to boost the power gain. For instance, the narrow-band internet-of-things standard allows for up to 2048 repetitions of a $1/3$ rate turbo code to meet the coverage requirement [14].

Recently, there has been remarkable advancements in the field of finite-blocklength information theory which outlines the attainable and converse bounds for the maximum channel code rate that can be achieved at a particular blocklength and error probability [15]. Despite this progress, designing an optimal code in finite-blocklength remains a formidable task due to the non-asymptotic behavior of codes in a finite-dimensional space. In a breakthrough, Arıkan, the inventor of polar codes [16], introduced a new family of polar codes named *polarization-adjusted convolutional (PAC) codes* [17]. Unlike the polar codes which are serially concatenated with a cyclic redundancy check (CRC) code used in 5G NR [18], PAC codes adopt a convolutional transform with a proper rate-profiling. PAC codes equipped with a tree-search sequential decoder have been shown to closely achieve the finite-length error probability lower bound, known as the meta-converse bound [15]. However, the sequential decoders (such as Fano or stack decoders) require a prohibitively high computational complexity at low signal-to-noise ratios (SNRs). More critically, the decoding latency of these decoders is unpredictable

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due to their inherent sequential nature, making them unsuitable for low-latency communication applications.

In this paper, we propose a novel approach to the design of low-rate codes for URLLC. By leveraging the advantages of sparsity in low-rate code design, we introduce a new class of codes called block orthogonal sparse superposition (BOSS) codes. Through the paper, we show that our BOSS code can achieve the meta-converse bound within one dB in the short-blocklength regime with a fast low-complexity decoder.

B. Related Work

Sparse superposition codes, or sparse regression codes (SAPRCs), are a joint modulation and coding technique introduced by Joseph and Barron [19]. Unlike traditional coded modulation techniques [20], [21], SPARCs construct codewords by multiplying a dictionary matrix with a sparse message vector, subject to a block sparsity constraint. With the use of Gaussian random dictionary matrices having independent and identically distributed (IID) entries for encoding and least squares decoding, SPARCs have demonstrated the capability of achieving any fixed rate smaller than the Gaussian channel capacity, as the code length tends to infinity [19].

Designing low-complexity decoders with performance comparable to the computationally-intractable maximum likelihood (ML) decoder in [19] is essential to make SPARCs practical [22], [23], [24], [25]. An efficient decoding algorithm called adaptive successive decoding was first introduced in [23], and its soft-decision variant was proposed in [22]. Decoding SPARCs with U sections can be viewed as a multi-user detection problem in a Gaussian multiple-access channel with U users and a sum-power constraint. The adaptive successive decoder leverages successive interference cancellation and a proper power allocation strategy. Compressed sensing-based decoding algorithms have also been explored as efficient alternatives due to the connection between SPARCs and compressed sensing [26]. The decoding problem can be viewed as sparse signal recovery from noisy measurements with a certain sparsity structure. Approximate message passing (AMP) [27], widely used in sparse recovery problems, has been proposed as a computationally-efficient decoding method for SPARCs [24], [25]. The AMP decoder's performance can be analyzed per iteration through its state evolution property [27], and its finite length performance is better than the original successive decoder.

The above works mainly focus on the asymptotic performance of SPARCs in a large system limit in order to prove the code's achievability of the Shannon capacity. Recently, efforts have been made to improve the finite-length performance of SPARCs. The AMP decoder performs heavy matrix-vector multiplications at each iteration. In order to reduce the complexity of AMP decoding, the Hadamard-based dictionary construction has been employed [24], [25], [28]. However, this requires sub-sampling of rows from a randomly permuted Hadamard matrix to reserve the statistical properties of the original Gaussian random dictionary matrix. Moreover, since each obtained section is not necessarily a square matrix, a message to be estimated needs to be prepended with zeros and then sub-sampled to leverage the fast Walsh-Hadamard transform each time. In [28], several practical techniques were proposed to improve the finite-length performance of SPARCs with AMP. These include an iterative power allocation algorithm and online method to compute an essential decoding parameter. In spite of these provisions, AMP is still sub-optimal particularly over small matrices, which critically bottlenecks the performance of SPARCs in the short blocklength regime (a few hundreds), a main target of this paper.

Perotti et al. [29] took a different approach and proposed to construct a dictionary matrix as a concatenation of matrices that are nearly orthogonal to each other, referred to as quasi-orthogonal sparse superposition codes (QO-SPARCs). QO-SPARCs based on Zadoff-Chu and Kerdock bent sequences were shown to perform better than polar codes with successive cancellation list (SCL) decoding in certain short-blocklength scenarios. However, constructing a QO-SPARC dictionary for each code rate and bloklength requires high computational complexity, and there exists a strict limit on the possible number of near-orthogonal matrices when using either of the two design sequences. Furthermore, the iterative decoding algorithm, called as belief propagation successive interference cancellation, suffers from high decoding complexity and latency, which cannot satisfy the stringent requirements of URLLC.

C. Contributions

The major contributions of this paper are summarized as follows:

- Our major contribution is the introduction of BOSS codes, a novel class of sparse superposition codes. Unlike SPARCs which utilize a randomly generated dictionary matrix for encoding, BOSS codes use a structured matrix created by concatenating G unitary matrices of size M by M. These unitary matrices (such as Fourier, Walsh-Hadamard, Haar, and discrete cosine matrices) can undergo fast unitary transforms for efficient encoding and decoding. The encoding process involves multiplying the structured dictionary matrix with a sparse message vector generated through a sequential bit-mapping strategy. This strategy maps the information bits into the positions and coefficients of sparse sub-message vectors using the non-selected column indices from previous sub-message vectors.
- We introduce a highly efficient and low-complexity decoder for BOSS codes, known as the two-stage MAP decoder, which provides near ML decoding performance. In the first stage, the decoder performs G parallel unitary transforms on the received signal with a complexity of $\mathcal{O}(GM\log(M))$. Then, each transformed signal is decoded independently using element-wise MAP decoding with successive support set cancellation, which has a complexity of $\mathcal{O}(KM)$. In the second stage, the decoder identifies the unitary matrix index through minimum distance detection, with a linear complexity of $\mathcal{O}(G)$. For two-layered BOSS codes, the most relevant design for low-rate codes, a simple ordered statistics (OS) decoder

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with $\mathcal{O}(K\log(M))$ linear complexity in the blocklength is optimal for the first stage decoding. Our two-stage decoder has the advantage of being easily implemented in parallel, which is crucial for low-latency communications.

- We provide an exact expression for the block error rates (BLERs) of a single-layered BOSS code using a two-stage decoding method. Our analytical expression takes into account the code blocklength, code rate, and SNR as the relevant code parameters. The results of our analysis demonstrate how the code performance varies based on these code design parameters. Our analytical expression for BLER is useful for predicting the minimum required SNR to achieve an extremely low target BLER of less than 10^{-8}, which is difficult to achieve even through computer simulations. Our findings also verify that our decoding method does not have an error-floor as the SNR increases.
- To enhance coding performances, we also put forth a CRC-aided BOSS code, called a CA-BOSS code, which is constructed by serially concatenating a BOSS code with a CRC code as an outer code. To decode the CA-BOSS code efficiently, we propose a list decoding method. The method distinguishes itself from the two-stage MAP decoder by first identifying a set of S codewords with high reliability and then validating if they meet the CRC constraints. In the second stage, it performs minimum distance detection for the CRC-valid codewords. The decoding complexity of the list decoder does not significantly increase the overall complexity, adding only a linear factor S, $\mathcal{O}(SK\log(M))$, to the OS decoder complexity. Our simulations show that the proposed CA-BOSS code outperforms the CA-polar code with SCL decoding for short blocklengths and low code rates. Furthermore, our code can achieve the finite-blocklength capacity within one dB with efficient encoding and decoding with a complexity order of $\mathcal{O}(GM\log(M))$.

The remainder of this paper is organized as follows. In Section II, we present an encoding technique for BOSS codes. In the next section, Section III, we introduce the approximate MAP decoder, which facilitates rapid decoding. In Section IV, we present an analytical expression for the BLER of single-layered BOSS codes when our proposed decoder is implemented. To validate our findings, we present numerical results in Section V. The final section, Section VI, concludes the paper by offering potential avenues for future research and extension.

II. Block Orthogonal Sparse Superposition Coding

In this section, we present a novel encoding strategy called successive orthogonal encoding. To augment understanding of BOSS coding, we introduce some useful properties and remarks.

A. Preliminaries

Before describing the code construction process, we introduce some notations and definitions used in this paper.

1) Additive White Gaussian Noise (AWGN) Channel: We mainly investigate a transmission of codewords over the AWGN channel. We consider the real AWGN channel, but the extension to the complex system is relatively simple. The blocklength, M, of the code is defined as an integer in $\mathbb{Z}^{+}$, while the rate, R, is defined as a positive real number in $\mathbb{R}^{+}$. During the transmission of a codeword, $\mathbf{c}$, which is a vector in $\mathbb{R}^{M}$, the received vector, $\mathbf{y}$, in $\mathbb{R}^{M}$ is obtained through the AWGN channel as

$$\mathbf{y} = \mathbf{c} + \mathbf{v}, \tag{1}$$

where $\mathbf{v}$ is an M-dimensional zero-mean Gaussian noise vector of variance σ_v^2, i.e., $\mathbf{v} \sim \mathcal{N}(\mathbf{0}_M, \sigma_v^2\mathbf{I}_M)$.

2) Dictionary and Sparse Message Vector: A BOSS code is defined by a dictionary matrix $\mathbf{A}$ of dimension $M \times N$, where $N \in \mathbb{Z}^{+}$ is the length of a sparse message vector. The dictionary matrix $\mathbf{A}$ is constructed as a concatenation of $G \in \mathbb{Z}^{+}$ unitary matrices $\mathbf{U}_g \in \mathbb{R}^{M\times M}$:

$$\mathbf{A} = \begin{bmatrix}\mathbf{U}_1 & \mathbf{U}_2 & \cdots & \mathbf{U}_G\end{bmatrix}. \tag{2}$$

It is worthwhile to note that any random unitary matrices can be used for encoding; yet, a set of special unitary matrices are preferred for efficient encoding and decoding. For instance, the encoder uses a Walsh-Hadamard matrix for $\mathbf{U}_1$. By taking a row permutation to $\mathbf{U}_1$, it is possible to construct other unitary matrices as

$$\mathbf{U}_g = \mathbf{P}_g\mathbf{U}_1, \tag{3}$$

where $\mathbf{P}_g \in \{0,1\}^{M\times M}$ is the gth permutation matrix. To make the sub-matrices distinguishable, the encoder should choose $G-1$ distinct permutation matrices, i.e., $\mathbf{P}_g \neq \mathbf{P}_{g'}$ for $g, g' \in \{2,3,\ldots,G\}$.

A BOSS codeword $\mathbf{c}$ is obtained as a matrix-vector multiplication of the dictionary matrix and a message vector $\mathbf{x} \in \mathbb{R}^{N}$: $\mathbf{c} = \mathbf{A}\mathbf{x}$. The message vector is sparse such that it only has K non-zero coefficients, i.e., $\|\mathbf{x}\|_0 = |\operatorname{supp}(\mathbf{x})| = K(\ll N)$. Here, the sparse message vector $\mathbf{x}$ is generated as a superposition of L sparse message vectors:

$$\mathbf{x} = \sum_{\ell=1}^{L}\mathbf{x}^{(\ell)}, \tag{4}$$

where $\mathbf{x}^{(\ell)} \in \mathbb{R}^{N}$ denotes the ℓth layered sparse sub-message vector with K_ℓ non-zero coefficients, i.e., $\|\mathbf{x}^{(\ell)}\|_0 = K_\ell$. The sparsity levels K_ℓ's satisfy the total sum constraint, $\sum_{\ell=1}^{L}K_\ell = K$. The encoder assigns different signal levels for the non-zero elements in $\mathbf{x}^{(\ell)}$. Let J_ℓ be a non-zero alphabet size of $\mathbf{x}_\ell$. The constellation for the ℓth sub-message vector is $\mathcal{A}_\ell = \{\alpha_{\ell,1}, \alpha_{\ell,2}, \ldots, \alpha_{\ell,J_\ell}\}$ such as a pulse-amplitude modulation (PAM) set. For simplicity in the decoding process, the signal level sets for distinct sub-message vectors are made to be disjoint, i.e., $\mathcal{A}_j \cap \mathcal{A}_k = \emptyset$ for $j \neq k \in [L]$, where $[L] \triangleq \{1,2,\ldots,L\}$. We also denote by $\mathbf{x}_g \in \mathbb{R}^{M}$ a segment of $\mathbf{x} \in \mathbb{R}^{N}$ that corresponds to a gth unitary matrix $\mathbf{U}_g$, i.e,

$$\mathbf{x} = \begin{bmatrix}\mathbf{x}_1^{\mathsf{T}} & \cdots & \mathbf{x}_g^{\mathsf{T}} & \cdots & \mathbf{x}_G^{\mathsf{T}}\end{bmatrix}^{\mathsf{T}}. \tag{5}$$

Then, $\mathbf{x}_g$ can also be represented as a superposition of corresponding segments of each ℓth layered sub-message vector $\mathbf{x}^{(\ell)}$: $\mathbf{x}_g = \sum_{\ell=1}^{L}\mathbf{x}_g^{(\ell)}$.

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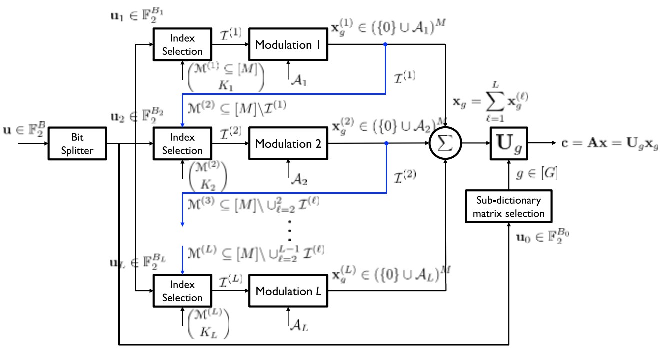

Fig. 1. Proposed sequential encoder structure for the BOSS code construction.

B. Sequential Bit Mapping

Fig. 1 illustrates the proposed sequential encoding scheme which performs the following operations.

- **Step 1 (Block selection):** The encoder uses the first $B_0 = \lfloor \log_2(G) \rfloor$ bits to select a sub-dictionary matrix index g: among G segments of $\mathbf{x}$, only $\mathbf{x}_g$ has K non-zero elements, i.e., $\|\mathbf{x}\|_0 = \|\mathbf{x}_g\|_0 = K$.
- **Step 2 (Non-zero index configuration):** The encoder selects K_1 columns of $\mathbf{U}_g$ by using the next $B_{1,\text{loc}} = \left\lfloor \log_2 \left(\binom{|\mathcal{M}^{(1)}|}{K_1} \right) \right\rfloor$ bits. Here, $\mathcal{M}^{(1)} \subseteq [M]$ is a candidate set of possible non-zero index locations, whose size is pre-determined. The selected columns correspond to the indices of non-zero elements of $\mathbf{x}_g^{(1)}$.
- **Step 3 (Non-zero value assignment):** The following $B_{1,\text{val}} = \left\lfloor \log_2 \left(|\mathcal{A}_1|^{K_1} \right) \right\rfloor$ bits are mapped into non-zero values. The encoder selects K_1 values uniformly at random from $\mathcal{A}_1$ and assigns them to the non-zero indices selected at the previous step. As a result, the gth segment of the first layer sub-vector $\mathbf{x}^{(1)}$ is obtained as $\mathbf{x}_g^{(1)} \in (\{0\} \cup \mathcal{A}_1)^M$. The corresponding first layer sub-codeword is given by $\mathbf{c}^{(1)} = \mathbf{A}\mathbf{x}^{(1)} = \mathbf{U}_g\mathbf{x}_g^{(1)}$.

We define the support of $\mathbf{x}_g^{(1)}$, i.e., a set of non-zero index locations, as $\mathcal{I}^{(1)} = \{m \in \mathcal{M}^{(1)} | x_{g,m}^{(1)} \in \mathcal{A}_1\}$. Previously chosen indices are excluded to avoid duplicate uses of locations. Utilizing the support information $\mathcal{I}^{(1)}$, the encoder determines a candidate set for the second layer as $\mathcal{M}^{(2)} \subseteq [M] \backslash \mathcal{I}^{(1)}$. The encoder then repeats the Step 2 and 3 for L layers and successively maps information bits in the same fashion. Note that the sparsity degree K_ℓ; signal level set $\mathcal{A}_\ell$; and number of bits for non-zero index selection $B_{\ell,\text{loc}}$ and value assignment $B_{\ell,\text{val}}$ per layer are pre-configured as code parameters and shared between the transmitter and receiver. A high-level description of the sequential BOSS encoding algorithm is given in Algorithm 1.

The crux of this sequential encoding strategy is that previous non-zero indices are never included in the subsequent candidate sets. Thanks to the orthogonal construction, the support of $\mathbf{x}_g^{(\ell)}$, $\mathcal{I}^{(\ell)}$, is mutually exclusive with the union of the supports of $\mathbf{x}_g^{(j)}$ for $j \in \{1, \dots, \ell-1\}$, i.e., $\mathcal{I}^{(\ell)} \cap \left(\cup_{j=1}^{\ell-1} \mathcal{I}^{(j)} \right)$. Given that index support sets of distinct layers are non-overlapping, a BOSS codeword can be represented as a superposition of L sub-codeword vectors, each of which is a linear combination of a subset of column vectors of $\mathbf{U}_g$:

$$\mathbf{c} = \sum_{\ell=1}^{L} \mathbf{c}^{(\ell)} = \sum_{\ell=1}^{L} \mathbf{A}\mathbf{x}^{(\ell)} = \sum_{\ell=1}^{L} \mathbf{U}_g \mathbf{x}_g^{(\ell)}. \tag{6}$$

Algorithm 1 Sequential BOSS Encoding

Input: input bit-stream $\mathbf{u}$ of length $B = B_0 + \sum_{\ell=1}^{L} B_\ell = B_0 + \sum_{\ell=1}^{L} (B_{\ell,\text{loc}} + B_{\ell,\text{val}})$, sparsity levels $\{K_\ell\}_{\ell \in [L]}$ with $\sum_{\ell=1}^{L} K_\ell = K$, and set of constellations $\{\mathcal{A}_\ell\}_{\ell \in [L]}$

Output: sparse message vector $\mathbf{x} \in \mathbb{R}^N$ with $\|\mathbf{x}\|_0 = K$

Use the first $B_0 = \lfloor \log_2(G) \rfloor$ bits to select a sub-dictionary matrix g

for $\ell = 1, \dots, L$ **do**

  Partition $\mathbf{u}_\ell \in \mathbb{F}_2^{B_\ell}$ into $\mathbf{u}_{\ell,\text{loc}} \in \mathbb{F}_2^{B_{\ell,\text{loc}}}$ and $\mathbf{u}_{\ell,\text{val}} \in \mathbb{F}_2^{B_{\ell,\text{val}}}$

  Configure $\mathcal{M}^{(\ell)} \subseteq [M] \backslash \left(\cup_{j=1}^{\ell-1} \mathcal{I}^{(i)} \right)$

  Use $B_{\ell,\text{loc}}$ bits to select K_ℓ column indices of $\mathbf{U}_g$ from $\mathcal{M}^{(\ell)}$

  Use $B_{\ell,\text{val}}$ bits to select K_ℓ values uniformly at random from $\mathcal{A}_\ell$

  Assign the selected values to the corresponding indices of $\mathbf{x}_g^{(\ell)}$

  /* $\mathbf{x}_g^{(\ell)} \in (\{0\} \cup \mathcal{A}_\ell)^M$ /*

  Define $\mathcal{I}^{(\ell)} = \{m \in \mathcal{M}^{(\ell)} | x_{g,m}^{(\ell)} \in \mathcal{A}_\ell\}$

  /* Support of $\mathbf{x}_g^{(\ell)}$ /*

return $\mathbf{x} = \sum_{\ell=1}^{L} \mathbf{x}^{(\ell)}$ /* $\mathbf{x}_g = \sum_{\ell=1}^{L} \mathbf{x}_g^{(\ell)}$ /*

C. Properties

To shed further light on the significance of our code construction method, we provide some properties.

1) Orthogonality: The most prominent property of the BOSS code is orthogonality between sub-codewords, i.e., $\langle \mathbf{c}^{(j)}, \mathbf{c}^{(k)} \rangle = 0$ for $j \neq k \in [L]$. This inherent orthogonal property helps develop a computationally efficient yet powerful decoder, which will be explained in Section III.

2) Decodability: In the absence of noise, a single-block, i.e., $G = 1$, BOSS code is uniquely decodable since $\mathcal{A}_j \cap \mathcal{A}_k = \emptyset$ for $j \neq k \in [L]$. Suppose $\mathbf{A} = \mathbf{I}_M$, i.e., $\mathbf{c} = \mathbf{x}$. This is true because a decoder is able to distinguish the ℓth sub-codeword $\mathbf{x}^{(\ell)}$ from $\mathbf{x} = \sum_{j=1}^{L} \mathbf{x}^{(j)}$, provided $x_m^{(\ell)} \notin \cup_{j \neq \ell} \mathcal{A}_j$, $\forall m \in [M]$. The decoder, then, performs de-mapping from $\mathbf{x}^{(\ell)}$ to $\mathbf{u}_\ell$, a stream of $(B_{\ell,\text{loc}} + B_{\ell,\text{val}})$ information bits.

3) Code Rate: At the ℓth layer, the encoder is allowed the leeway to choose K_ℓ indices from $\mathcal{M}^{(\ell)}$ and then K_ℓ values from $\mathcal{A}_\ell$. That is, a total of $B_{\ell,\text{loc}} + B_{\ell,\text{val}}$ bits are mapped into the locations and values of non-zero entries of $\mathbf{x}_g^{(\ell)}$, respectively. Then, the ℓth layer's sub-codeword $\mathbf{c}^{(\ell)} = \mathbf{A}\mathbf{x}^{(\ell)}$ conveys $B_\ell = B_{\ell,\text{loc}} + B_{\ell,\text{val}} = \left\lfloor \log_2 \left(\binom{|\mathcal{M}^{(\ell)}|}{K_\ell} \right) \right\rfloor + \lfloor K_\ell \log_2(|\mathcal{A}_\ell|) \rfloor$ information bits. Taking account of B_0 bits mapped into the block index, the rate of the BOSS code is

$$R = \frac{\lfloor \log_2(G) \rfloor + \sum_{\ell=1}^{L} \left(\left\lfloor \log_2 \left(\binom{|\mathcal{M}^{(\ell)}|}{K_\ell} \right) \right\rfloor + \lfloor K_\ell \log_2(|\mathcal{A}_\ell|) \rfloor \right)}{M}. \tag{7}$$

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For a symmetric case in which $|\mathcal{M}^{(\ell)}| = \mathcal{M}(< M)$, $K_\ell = \mathcal{K}$, and $J_\ell = 1$ for all $\ell \in [L]$, the code rate is simplified to $R = \frac{\lfloor \log_2(G) \rfloor + L \lfloor \log_2(\binom{\mathcal{M}}{\mathcal{K}}) \rfloor}{M}$. For a fixed blocklength M, the proposed encoding scheme can construct codes of various rates by appropriately tuning multiple design parameters: the number of blocks G, the layer depth L, the number of non-zeros per layer K_ℓ, and the size of non-zero alphabets J_ℓ. This flexibility in coding rate is a salient feature of BOSS coding and attests to its applicability to URLLC use-cases.

4) Average Transmit Power: Without loss of generality, we assume that the encoder has chosen the very first sub-dictionary matrix, i.e., $\|\mathbf{x}\|_0 = \|\mathbf{x}_1\|_0 = K$. The ℓth layer's vector $\mathbf{x}_1^{(\ell)}$ has K_ℓ elements drawn from $\mathcal{A}_\ell$, and the signal energy associated with few non-zero entries is distributed by $\mathbf{U}_1$. Since the norm is preserved under unitary transformation, the average power of a sub-codeword $\mathbf{c}^{(\ell)}$ is given by

$$\begin{aligned}\mathbb{E}\{\|\mathbf{c}^{(\ell)}\|_2^2\} &= \mathbb{E}\{\|\mathbf{U}_1\mathbf{x}_1^{(\ell)}\|_2^2\} = \mathbb{E}\{\|\mathbf{x}_1^{(\ell)}\|_2^2\} \\ &= \frac{K_\ell \frac{\sum_{j=1}^{J_\ell} \alpha_{\ell,j}^2}{J_\ell}}{M}. \end{aligned} \tag{8}$$

Since all sub-codeword vectors are orthogonal, the average transmit power becomes

$$E_{\mathrm{s}} = \mathbb{E}\{\|\mathbf{A}\mathbf{x}\|_2^2\} = \sum_{\ell=1}^{L} \mathbb{E}\{\|\mathbf{x}_1^{(\ell)}\|_2^2\} = \sum_{\ell=1}^{L} \frac{K_\ell \sum_{j=1}^{J_\ell} \alpha_{\ell,j}^2}{J_\ell M}. \tag{9}$$

Example: For ease of exposition, let us consider a BOSS code with $[G, M, L] = [8, 64, 2]$, $K_1 = K_2 = 1$, and singleton PAM alphabets $\mathcal{A}_1 = \{1\}$ and $\mathcal{A}_2 = \{-1\}$. The first $B_0 = \lfloor \log_2(8) \rfloor = 3$ bits are encoded into a block index $g \in [8]$. The encoder maps the following $B_1 = \lfloor \log_2\left(\binom{64}{1}\right) \rfloor = 6$ bits into choosing a column index $i_1 \in [64]$. Without loss of generality, suppose $g = 1$ and $i_1 = 1$. Then, $\mathbf{x}_1^{(1)}$ has a single non-zero coefficient, i.e., $x_1^{(1)} = 1$. The encoder chooses a second position i_2 from $\mathcal{M}^{(2)} \subseteq [M] \backslash \{i_1\} = \{2, 3, \ldots, 64\}$ represented by the last $B_2 = \left\lfloor \log_2\left(\binom{63}{1}\right) \right\rfloor = 5$ bits based on the predefined mapping table, and assigns -1 to the selected entry. The codeword is then a linear combination of the selected columns of $\mathbf{U}_1$, i.e., $\mathbf{c} = \mathbf{U}_1\left(\mathbf{x}_1^{(1)} + \mathbf{x}_1^{(2)}\right) = \mathbf{u}_{1,i_1} - \mathbf{u}_{1,i_2}$, where $\mathbf{u}_{1,j}$ is a jth column vector of $\mathbf{U}_1$ for $j \in [M]$. A codeword delivers a total of $14 = (3 + 6 + 5)$ bits with 64 channels uses, i.e., $R = \frac{14}{64} = 0.21875$. The normalized average transmit power per channel use becomes $E_{\mathrm{s}} = \frac{1}{64} + \frac{1}{64} = \frac{1}{32}$.

D. Remarks

We provide some remarks to offer further insights into BOSS coding and highlight the difference with the existing coding and modulation methods.

Remark 1 (Joint Encoding): The block-index selection and first-layer bit-mapping can be combined into a single process. The encoder chooses a single column index from a fat dictionary matrix, i.e., $\binom{N}{1} = \binom{G}{1}\binom{M}{1}$, and the remaining $K_1 - 1$ indices are selected from a corresponding block. That is, putting the alphabet allocation aside,

$$\begin{aligned}&\log_2(G) + \log_2\left(\binom{M}{K_1}\right) \\ &\qquad = \log_2(N) + \log_2\left(\binom{M-1}{K_1-1} \cdot \frac{1}{K_1}\right). \end{aligned} \tag{10}$$

Remark 2 (Orthogonal Multiplexing for Multi-Layer Index Modulation Signals): The proposed coding scheme also generalizes the existing index modulation methods [30]. Consider BOSS encoding with $G = 1$ and $L = 1$. The resulting code is identical to the index modulation. Therefore, our coding scheme can be interpreted as an efficient orthogonal multiplexing method of multi-layer index (spatial) modulated signals [30], [31], [32].

III. Approximate MAP Decoder

This section presents an approximate MAP decoder featuring low decoding latency and complexity. We first explain the motivation of the approximate MAP decoder and present the two-stage decoding algorithm in the sequel.

A. Two-Stage MAP Approximation

Under the premise that a codeword $\mathbf{c}$ is generated using the gth sub-dictionary matrix in $\mathbf{A}$, the codeword can be written as

$$\mathbf{c} = \mathbf{A}\mathbf{x} = \mathbf{U}_g\mathbf{x}_g, \tag{11}$$

where the sub-message vector $\mathbf{x}_g = \sum_{\ell=1}^{L} \mathbf{x}_g^{(\ell)}$ is a member of the set

$$\mathcal{X}_g = \left\{ \mathbf{x}_g \in \mathbb{R}^M | \mathbf{x}_g \in \mathcal{A}^M, \|\mathbf{x}_g\|_0 = \sum_{\ell=1}^{L} K_\ell \right\}, \tag{12}$$

where $\mathcal{A} = \{0\} \bigcup (\cup_{\ell=1}^{L} \mathcal{A}_\ell)$. Note that $\mathcal{X}_g$ is different from the set of possible sparse message vectors defined as

$$\mathcal{X} = \left\{ \mathbf{x} \in \mathbb{R}^N | \mathbf{x} \in \mathcal{A}^N, \|\mathbf{x}\|_0 = \sum_{\ell=1}^{L} K_\ell, \sum_{g=1}^{G} \mathbf{1}_{\|\mathbf{x}_g\| \neq 0} = 1 \right\}. \tag{13}$$

Let $\mathcal{H}_g$ be a hypothesis that the transmitted codeword is constructed with a sub-dictionary matrix $\mathbf{U}_g$:

$$\mathcal{H}_g : \mathbf{c} = \mathbf{U}_g\mathbf{x}_g. \tag{14}$$

Then, the MAP decoding problem can be decomposed into two different sub-MAP tasks, i.e.,

$$\begin{aligned}\hat{\mathbf{x}}^{\mathrm{MAP}} &= \arg\max_{\mathbf{x} \in \mathcal{X}} \mathbb{P}(\mathbf{x}|\mathbf{y}) \\ &= \arg\max_{\mathbf{x}_g \in \mathcal{X}_g, g \in [G]} \mathbb{P}(\mathbf{x}_g, \mathcal{H}_g|\mathbf{y}) \\ &= \arg\max_{\mathbf{x}_g \in \mathcal{X}_g, g \in [G]} \mathbb{P}(\mathbf{x}_g|\mathbf{y}, \mathcal{H}_g)\mathbb{P}(\mathcal{H}_g|\mathbf{y}). \end{aligned} \tag{15}$$

Solving this MAP decoding problem requires the complexity order of $\mathcal{O}(|\mathcal{X}|)$, which is prohibitively high as the number of information bits increases. To diminish the decoding complexity, we present a two-stage MAP decoder. The key idea of the two-stage MAP decoder is to solve the MAP decoding

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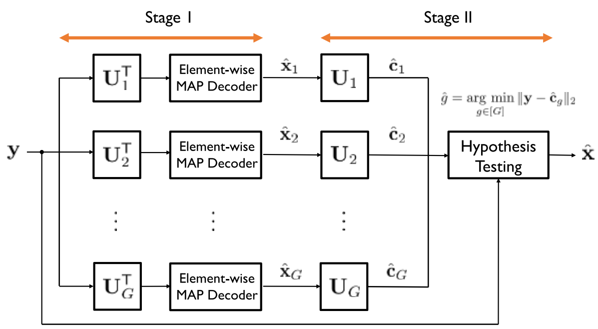

Fig. 2. Two-stage MAP decoder for BOSS codes in the AWGN channel.

problem in (15) independently for $\mathbf{x}_g \in \mathcal{X}_g$ and $g \in [G]$, respectively. To be specific, in the first stage, the decoder first identifies $\hat{\mathbf{x}}_g$ under $\mathcal{H}_g$

$$\hat{\mathbf{x}}_g = \arg\max_{\mathbf{x}_g \in \mathcal{X}_g} \mathbb{P}(\mathbf{x}_g|\mathbf{y}, \mathcal{H}_g). \tag{16}$$

Then, in the second stage, using the decoded $\hat{\mathbf{x}}_g$, it finds the block index $\hat{g}$:

$$\hat{g} = \arg\max_{g\in[G]} \mathbb{P}(\mathcal{H}_g|\mathbf{y}) = \arg\max_{g\in[G]} \mathbb{P}(\mathbf{c} = \mathbf{U}_g\hat{\mathbf{x}}_g|\mathbf{y}). \tag{17}$$

B. Two-Stage MAP Decoder

We explain our two-stage MAP decoding algorithm.

1) Sparse Message Vector Recovery: As illustrated in Fig. 2, in the first stage, the decoder aims at identifying $\mathbf{x}_g$ under each hypothesis $\mathcal{H}_g$ in parallel. To this end, the decoder simply takes a fast unitary transform to the received signal vector $\mathbf{y}$ and produces G transformed received vectors as

$$\mathbf{y}_g = \mathbf{U}_g^\mathsf{T}\mathbf{y} = \mathbf{x}_g + \tilde{\mathbf{v}}_g, \tag{18}$$

for $g \in [G]$, and $\tilde{\mathbf{v}}_g = \mathbf{U}_g^\mathsf{T}\mathbf{v}$ follows the identical distribution as $\mathbf{v}$ because the Gaussian distribution is invariant under the unitary transform. The complexity order to obtain $\mathbf{y}_g$ for $g \in [G]$ is $\mathcal{O}(GM\log(M))$ thanks to the fast transform for $\mathbf{U}_g^\mathsf{T} = \mathbf{U}_1^\mathsf{T}\mathbf{P}_g^\mathsf{T}$.

Now, the decoder solves the first sub-MAP decoding problem using $\mathbf{y}_g$:

$$\hat{\mathbf{x}}_g = \arg\max_{\mathbf{x}_g\in\mathcal{X}_g} \mathbb{P}\left(\mathbf{x}_g|\mathbf{y}, \mathcal{H}_g\right) = \arg\max_{\mathbf{x}_g\in\mathcal{X}_g} \mathbb{P}\left(\mathbf{x}_g|\mathbf{y}_g\right). \tag{19}$$

Exploiting the fact that the support of $\mathbf{x}_g$ is a union of mutual-exclusive index sets from L layers, we compute and factorize the log of joint a posteriori probability (APP) in (19):

$$\begin{aligned}\log \mathbb{P}(\mathbf{x}_g|\mathbf{y}_g) &= \sum_{\ell=1}^{L} \log \mathbb{P}(\mathbf{x}_g^{(\ell)}|\mathbf{y}_g, \mathbf{x}_g^{(\ell-1)}, \ldots, \mathbf{x}_g^{(2)}, \mathbf{x}_g^{(1)}) \\ &= \sum_{\ell=1}^{L} \log \mathbb{P}(\mathbf{x}_g^{(\ell)}|\mathbf{y}_g, \mathcal{I}_g^{(\ell-1)}, \ldots, \mathcal{I}_g^{(2)}, \mathcal{I}_g^{(1)}),\end{aligned} \tag{20}$$

where the second equality follows from that previous non-zero supports, $\mathcal{I}_g^{(\ell-1)}, \ldots, \mathcal{I}_g^{(2)}, \mathcal{I}_g^{(1)}$, provide sufficient information to decode $\mathbf{x}_g^{(\ell)}$. Hence, the decoder shall leverage side information obtained in the preceding layers to recover $\mathbf{x}_g^{(\ell)}$ and update the conditional likelihoods accordingly. This motivates to design a decoding algorithm based on successive support set cancellation. From the Bayes's rule as in [33] and [34], the ℓth layer's log likelihood in (20) is reformulated as

$$\begin{aligned}&\log \mathbb{P}(\mathbf{x}_g^{(\ell)}|\mathbf{y}_g, \hat{\mathcal{I}}_g^{(\ell-1)}, \ldots, \hat{\mathcal{I}}_g^{(2)}, \hat{\mathcal{I}}_g^{(1)}) \\ &= \frac{1}{C}\sum_{m\in\hat{\mathcal{M}}_g^{(\ell)}} \log \frac{\mathbb{P}(y_{g,m}|x_{g,m}^{(\ell)})\mathbb{P}(x_{g,m}^{(\ell)})}{\mathbb{P}(y_{g,m})} \mathbf{1}_{\{\|\mathbf{x}_g^{(\ell)}\|_0 = K_\ell\}},\end{aligned} \tag{21}$$

where $C \in \mathbb{R}^+$ is a normalizing constant, and a candidate set estimate $\hat{\mathcal{M}}_g^{(\ell)}$ is defined as

$$\hat{\mathcal{M}}_g^{(\ell)} \subseteq [M] \backslash \cup_{j=1}^{\ell-1} \hat{\mathcal{I}}_g^{(j)}. \tag{22}$$

Here and hereinafter, we omit support estimates obtained beforehand for notational simplicity in (21), as they are already manifested in $\hat{\mathcal{M}}_g^{(\ell)}$. The ℓth layer candidate set estimate in (22) is updated at each iteration so that prior information on previous guesses is properly incorporated into subsequent decoding processes. Note that estimates in (22) may differ by hypothesis $\mathcal{H}_g$, but they all feature the same cardinality predefined in encoding. The element-wise conditional likelihood function in (21) is given by

$$\mathbb{P}\left(y_{g,m}|x_{g,m}^{(\ell)}\right) = \begin{cases} \dfrac{1}{J_\ell}\displaystyle\sum_{j=1}^{J_\ell} \frac{1}{\sqrt{2\pi\sigma_v^2}} e^{-\frac{|y_{g,m}-\alpha_{\ell,j}|^2}{2\sigma_v^2}} & \text{if } x_{g,m}^{(\ell)} \in \mathcal{A}_\ell \\ \dfrac{1}{\sqrt{2\pi\sigma_v^2}} e^{-\frac{|y_{g,m}|^2}{2\sigma_v^2}} & \text{else.} \end{cases} \tag{23}$$

As values of $\mathcal{A}_\ell$ are uniformly allocated to non-zero entries of $\mathbf{x}_g^{(\ell)}$, a prior distribution of $x_{g,m}^{(\ell)}$ is given by

$$\mathbb{P}(x_{g,m}^{(\ell)}) = \begin{cases} p_{g,m}^{(\ell)}, & \text{if } x_{g,m}^{(\ell)} \in \mathcal{A}_\ell \\ 1 - p_{g,m}^{(\ell)}, & \text{if } x_{g,m}^{(\ell)} = 0, \end{cases} \tag{24}$$

where $p_{g,m}^{(\ell)}$ is the probability of $x_{g,m}^{(\ell)}$ being a non-zero element given by

$$p_{g,m}^{(\ell)} = \frac{K_\ell}{|\mathcal{M}^{(\ell)}|}. \tag{25}$$

Invoking (23) and (24), we have the marginal distribution of $y_{g,m}$:

$$\begin{aligned}\mathbb{P}(y_{g,m}) &= \mathbb{P}(y_{g,m}|x_{g,m}^{(\ell)} \in \mathcal{A}_\ell)\mathbb{P}(x_{g,m}^{(\ell)} \in \mathcal{A}_\ell) \\ &\quad + \mathbb{P}(y_{g,m}|x_{g,m}^{(\ell)} = 0)\mathbb{P}(x_{g,m}^{(\ell)} = 0) \\ &= \frac{1}{J_\ell}\sum_{j=1}^{J_\ell} \frac{1}{\sqrt{2\pi\sigma_v^2}} e^{-\frac{|y_{g,m}-\alpha_{\ell,j}|^2}{2\sigma_v^2}} p_{g,m}^{(\ell)} \\ &\quad + \frac{1}{\sqrt{2\pi\sigma_v^2}} e^{-\frac{|y_{g,m}|^2}{2\sigma_v^2}} (1 - p_{g,m}^{(\ell)}).\end{aligned} \tag{26}$$

Note that recovery of a sparse vector $\mathbf{x}_g^{(\ell)}$ is equivalent to identification of its non-zero element locations such that paired columns of $\mathbf{U}_g$ constitute a sub-codeword $\mathbf{c}^{(\ell)}$. For every index $m \in \hat{\mathcal{M}}_g^{(\ell)}$, the decoder computes its probability of being an

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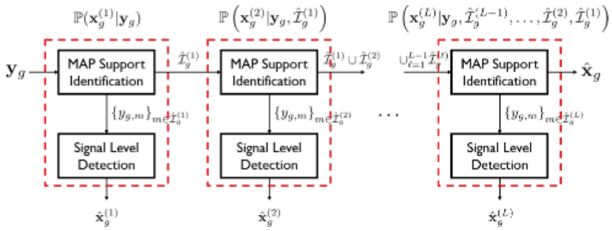

Fig. 3. Decoding of a sparse message vector under hypothesis $\mathcal{H}_g$.

element of $\mathcal{I}^{(\ell)}$, i.e. a likelihood of $x_{g,m}^{(\ell)}$ taking on value from $\mathcal{A}_\ell$ conditioned on $y_{g,m}$, which is given in (27), shown at the bottom of the page. The decoder sorts $\hat{\mathcal{M}}_g^{(\ell)}$ by the MAP metric in (27), which is monotone increasing with respect to $y_{g,m}$. To satisfy the sparsity requirement $\mathbf{1}_{\{\|\mathbf{x}_g^{(\ell)}\|_0 = K_\ell\}}$ in (21), the decoder selects K_ℓ indices that have the highest likelihoods and generates an ordered support estimate $\hat{\mathcal{I}}_g^{(\ell)}$:

$$\hat{\mathcal{I}}_g^{(\ell)} = \left\{\hat{i}_{g,1}^{(\ell)}, \hat{i}_{g,2}^{(\ell)}, \ldots, \hat{i}_{g,K_\ell}^{(\ell)}\right\}, \tag{28}$$

where $\mathbb{P}(\hat{i}_{g,j}^{(\ell)} \in \mathcal{I}^{(\ell)}|y_{g,j}) \geq \mathbb{P}(\hat{i}_{g,k}^{(\ell)} \in \mathcal{I}^{(\ell)}|y_{g,k})$ for $j < k$. Once the layer support estimate is determined, the decoder performs another MAP estimation to identify the signal levels of $x_{g,m}^{(\ell)}$ for $m \in \hat{\mathcal{I}}_g^{(\ell)}$. In particular, this simplifies to the minimum Euclidean distance decoding:

$$\hat{x}_{g,m}^{(\ell)} = \underset{\alpha_{\ell,j} \in \mathcal{A}_\ell}{\arg\min} |y_{g,m} - \alpha_{\ell,j}|. \tag{29}$$

After L iterations, we obtain $\hat{\mathbf{x}}_g$ whose support is $\hat{\mathcal{I}}_g = \bigcup_{\ell=1}^{L} \hat{\mathcal{I}}_g^{(\ell)}$. Fig. 3 describes the first stage under hypothesis $\mathcal{H}_g$.

2) The Block Index Recovery: Using $\hat{\mathbf{x}}_g$ for $g \in [G]$, in the second stage, the decoder performs the hypothesis testing to identify a true block index g. For every $\hat{\mathbf{x}}_g$, the decoder generates a tentative codeword as $\hat{\mathbf{c}}_g = \mathbf{U}_g \hat{\mathbf{x}}_g$. Since $\mathbf{v}$ is the Gaussian white noise vector, and the encoder has selected the block index uniformly from $[G]$, the second sub-MAP decoding problem is reduced to seeking $\hat{\mathbf{c}}_g$ that is closest to $\mathbf{y}$:

$$\hat{g} = \underset{g \in [G]}{\arg\max}\, \mathbb{P}(\mathcal{H}_g|\mathbf{y}) = \underset{g \in [G]}{\arg\min} \|\mathbf{y} - \hat{\mathbf{c}}_g\|_2. \tag{30}$$

As a result, the decoder obtains the estimate of a sparse message vector of dimension N with $K = \sum_{\ell=1}^{L} K_\ell$ non-zero entries, all located in the $\hat{g}$th segment, as

$$\hat{\mathbf{x}}^{\text{MAP}} = \begin{bmatrix} \mathbf{0}_M^\mathsf{T} & \cdots & \hat{\mathbf{x}}_{\hat{g}}^\mathsf{T} & \cdots & \mathbf{0}_M^\mathsf{T} \end{bmatrix}^\mathsf{T}. \tag{31}$$

Remark 3 (Decoding Complexity): The proposed two-stage MAP decoder has a sub-quadratic decoding complexity of $\mathcal{O}(GM \log(M))$ in terms of the blocklength. The decoder first computes $\mathbf{y}_g$ under the hypothesis of $\mathcal{H}_g$ by performing a permutation using $\mathbf{P}_g$ and a fast transform with $\mathbf{U}_1$, resulting in a complexity of $\mathcal{O}(GM \log(M))$, Then, the decoder calculates the APPs in (27) and selects the indices with the largest values to estimate the support, with a complexity of $\mathcal{O}\left(\left(M - \sum_{j=1}^{\ell-1} K_j\right) \log_2(K_\ell)\right)$. The decoder performs element-wise maximum likelihood signal level detection for each layer, taking $J_\ell K_\ell$ computations. The decoder repeats these steps for each sub-message estimate to obtain $\hat{\mathbf{x}}_g$. Finally, the decoder generates codeword candidates and compares their Euclidean distance with $\mathbf{y}$, which has a negligible contribution to the overall decoding complexity. The total decoding complexity is $\mathcal{O}(GM \log(M))$, which is sub-quadratic in the blocklength.

Remark 4 (Optimality of a Simple Ordered Statistics Decoder): We consider a two-layered BOSS code with uniform sparsity $K_1 = K_2$; $\mathcal{A}_1 = \{1\}$ and $\mathcal{A}_2 = \{-1\}$; and an arbitrary block size G. In this case, we show that the first procedure of the proposed two-stage MAP decoding algorithm is equivalent to a simple ordered statistics (OS) decoder. By plugging

$$\mathbb{P}(y_{g,m}|m \in \mathcal{I}^{(1)}) = \frac{1}{\sqrt{2\pi\sigma_v^2}} \exp\left(-\frac{(y_{g,m} - 1)^2}{2\sigma_v^2}\right) \tag{32}$$

and

$$\begin{aligned} &\mathbb{P}(y_{g,m}|m \notin \mathcal{I}^{(1)}) \\ &\quad = \mathbb{P}(y_{g,m}|m \in \mathcal{I}^{(2)})\mathbb{P}(m \in \mathcal{I}^{(2)}) \\ &\qquad + \mathbb{P}\left(y_{g,m}|m \notin (\mathcal{I}^{(1)} \cup \mathcal{I}^{(2)})\right)\mathbb{P}(m \notin (\mathcal{I}^{(1)} \cup \mathcal{I}^{(2)})) \\ &\quad = \frac{1}{\sqrt{2\pi\sigma_v^2}} \exp\left(-\frac{(y_{g,m} + 1)^2}{2\sigma_v^2}\right) \frac{K_1}{M - K_1} \\ &\qquad + \frac{1}{\sqrt{2\pi\sigma_v^2}} \exp\left(-\frac{y_{g,m}^2}{2\sigma_v^2}\right) \frac{M - 2K_1}{M} \end{aligned} \tag{33}$$

into (27), we obtain the log APP of an event $m \in \mathcal{I}^{(1)}$:

$$\begin{aligned} &\log \mathbb{P}(m \in \mathcal{I}^{(1)}|y_{g,m}) \\ &= \frac{1}{1 + \exp\left(-\frac{2y_{g,m}}{\sigma_v^2}\right) \frac{M}{M-K_1} + \exp\left(-\frac{2y_{g,m}-1}{2\sigma_v^2}\right) \frac{M-2K_1}{K_1}}. \end{aligned} \tag{34}$$

$\log \mathbb{P}(m \in \mathcal{I}^{(1)}|y_{g,m})$ is a monotonically increasing function of $y_{g,m}$, so we conclude that the support estimate $\hat{\mathcal{I}}_g^{(1)}$ is determined by the K_1 largest values in $\mathbf{y}_g$. Similarly, $\hat{\mathcal{I}}_g^{(2)}$

$$\begin{aligned} \log \mathbb{P}(m \in \mathcal{I}^{(\ell)}|y_{g,m}) &= \log \frac{\mathbb{P}(y_{g,m}|x_{g,m}^{(\ell)} \in \mathcal{A}_\ell)\mathbb{P}(x_{g,m}^{(\ell)} \in \mathcal{A}_\ell)}{\mathbb{P}(y_{g,m})} \\ &= \log \frac{\frac{1}{J_\ell}\sum_{j=1}^{J_\ell} \frac{1}{\sqrt{2\pi\sigma_v^2}} e^{-\frac{|y_{g,m}-\alpha_{\ell,j}|^2}{2\sigma_v^2}} p_{g,m}^{(\ell)}}{\frac{1}{J_\ell}\sum_{j=1}^{J_\ell} \frac{1}{\sqrt{2\pi\sigma_v^2}} e^{-\frac{|y_{g,m}-\alpha_{\ell,j}|^2}{2\sigma_v^2}} p_{g,m}^{(\ell)} + \frac{1}{\sqrt{2\pi\sigma_v^2}} e^{-\frac{|y_{g,m}|^2}{2\sigma_v^2}} (1 - p_{g,m}^{(\ell)})}. \end{aligned} \tag{27}$$

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is equivalent to a set of K_2 smallest entries in $\mathbf{y}_g$, except $y_{g,j}$ for $j \in \hat{\mathcal{I}}_g^{(1)}$.

This OS decoder is particularly useful for the fast decoding because the sorting algorithm requiring a complexity of $\mathcal{O}(\log(K_1+K_2)M)$ is sufficient to decode $\hat{\mathbf{x}}_g$ from $\mathbf{y}_g$. In our simulations, we shall use this OS decoder for implementation efficiency.

IV. Performance Analysis

In this section, we present an analytical expression for the BLERs of single-layered BOSS codes when decoded using the proposed two-stage MAP decoding. Our analysis provides an insight into how the BLER changes according to the blocklength M, the number of unitary matrices G, and $\frac{E_\mathrm{b}}{N_0}$. The following theorem is the main result of this section.

Theorem 1: Let $\mathcal{E}_1$ and $\mathcal{E}_2$ be the error events for the first and second stage decoding. The BLER of a single-layered BOSS code with $\mathcal{A} = \{1\}$ and rate $R = \frac{\lfloor \log_2(M)\rfloor + \lfloor \log_2(G)\rfloor}{M}$ is given by

$$P_{\mathsf{BLER}}(M, G, \sigma_v^2) = \mathbb{P}(\mathcal{E}_1) + \mathbb{P}(\mathcal{E}_2|\mathcal{E}_1^c)\mathbb{P}(\mathcal{E}_1^c), \tag{35}$$

where

$$\mathbb{P}(\mathcal{E}_1) = 1 - \frac{(M-1)}{\sqrt{2\pi\sigma_v^2}}\int_{-\infty}^{\infty} Q\left(\frac{y-1}{\sigma_v}\right) \times \left[1 - Q\left(\frac{y}{\sigma_v}\right)\right]^{M-2} e^{-\frac{y^2}{2\sigma_v^2}}\mathrm{d}y, \tag{36}$$

and

$$\mathbb{P}(\mathcal{E}_2|\mathcal{E}_1^c) = 1 - \left[\frac{\Gamma\left(\frac{M}{2}\right)}{\sqrt{\pi}\Gamma\left(\frac{M-1}{2}\right)} \times \int_{-1}^{1} Q\left(\frac{w-1}{\sqrt{2\sigma_v^2}}\right)(1-w^2)^{\frac{M-3}{2}}\mathrm{d}w\right]^{M(G-1)}. \tag{37}$$

Proof: See Appendix.

Although the BLER expression in Theorem 1 is integral form, it illuminates how the decoding performance of BOSS codes is affected by code parameters. Note that the noise power σ_v^2 is related to the energy per information bit to the noise power spectral density ratio $\frac{E_\mathrm{b}}{N_0}$ and code rate $R = \frac{\lfloor \log_2(M)\rfloor + \lfloor \log_2(G)\rfloor}{M}$ as

$$\sigma_v^2 = \frac{1}{M}\frac{1}{2R\frac{E_\mathrm{b}}{N_0}} = \frac{1}{2(\lfloor \log_2(M)\rfloor + \lfloor \log_2(G)\rfloor)\frac{E_\mathrm{b}}{N_0}}. \tag{38}$$

The relationship between the blocklength M, the number of unitary blocks G, and the normalized SNR $\frac{E_\mathrm{b}}{N_0}$ has a significant impact on the BLER of the BOSS code. Firstly, for a fixed $\frac{E_\mathrm{b}}{N_0}$ and G, it can be observed that as M increases, both $\mathbb{P}(\mathcal{E}_1)$ and $\mathbb{P}(\mathcal{E}_2)$ decrease. This is because the effective noise power σ_v^2 decreases with M, and $\mathbb{P}(\mathcal{E}_1)$ and $\mathbb{P}(\mathcal{E}_2)$ are decreasing functions with respect to M. This supports the notion that a larger blocklength in the BOSS code can improve its BLER performance.

Secondly, for a fixed M, the impact of increasing G on the BLERs is dependent on the noise power σ_v^2. When $\frac{E_\mathrm{b}}{N_0}$ is high, the second stage decoding error can vanish, meaning $\mathbb{P}(\mathcal{E}_2|\mathcal{E}_1^c) \to 0$ as $\sigma_v^2 \to 0$. This is surprising as increasing the code rate by using more G unitary blocks does not result in any second stage decoding errors, provided that $\frac{E_\mathrm{b}}{N_0}$ is high enough. In this scenario, the BLER is only limited by the first stage decoding error. On the other hand, when $\frac{E_\mathrm{b}}{N_0}$ is not high enough, increasing G will worsen the second stage decoding performance.

Lastly, for fixed G and M, it is clear that as $\frac{E_\mathrm{b}}{N_0}$ increases, the decoding error approaches zero. This confirms that the single-layer BOSS code with the proposed two-stage MAP decoder does not have error-floor effects, which is crucial for applications such as telesurgery and augmented reality where extremely low target BLERs ($<10^{-6}$) are required.

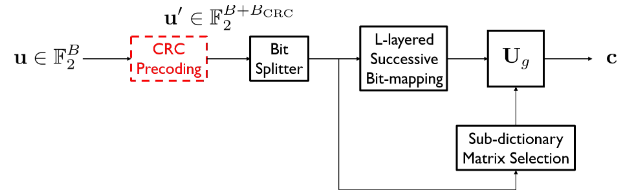

Fig. 4. Concatenated encoding of CA-BOSS codes.

V. CA-BOSS Codes

In this section, we introduce a novel approach to BOSS codes by implementing a serial concatenation scheme with a short CRC as the outer code. The resulting family of codes, referred to as CA-BOSS codes, significantly enhance the decoding performance through the use of a simple list decoder, while only slightly increasing the decoding complexity.

A. Concatenated Encoding

The proposed MAP-based decoding algorithm is prone to errors of the first stage decoding. To mitigate this problem, we incorporate CRC precoding. This process involves adding B_{CRC} CRC bits to the data block $\mathbf{u} \in \mathbb{F}_2^B$, forming a new block $\mathbf{u}' \in \mathbb{F}_2^{B+B_{\mathrm{CRC}}}$. This appended block is then processed through the original BOSS encoder, and the fractions are mapped into a dictionary block index and selected non-zero elements. The enhanced version of the encoder, referred to as the CA-BOSS encoder, is depicted in Fig. 4. The new addition to the original BOSS encoder is indicated in red.

B. List Decoder

We present a novel list decoding algorithm that leverages the two-stage MAP decoder and takes advantage of the error-detection capability of CRC. The decoder processes each $\mathcal{H}_g$ and produces S sub-message vector estimates, represented as $\{\hat{\mathbf{x}}_{g,1}, \hat{\mathbf{x}}_{g,2}, \ldots, \hat{\mathbf{x}}_{g,S}\}$. The generation of the list can vary, as demonstrated by an example of evaluating a two-layered BOSS code with $K_1 = K_2 = 1$, where the decoder selects the two most probable indices at each layer, resulting in a list of $S = 4$ combinations. The decoder then maps each provisional support $\hat{\mathcal{I}}_{g,s}$ and g to a bit sequence and performs a CRC test to determine the valid combination of the block and

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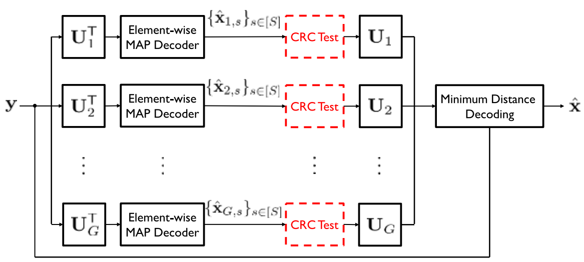

Fig. 5. The MAP-List decoder design of CA-BOSS codes.

non-zero elements. The surviving message vector estimates are transformed to the next stage and re-encoded into tentative codewords. The decoder finally selects the codeword candidate closest to $\mathbf{y}$, determining the block index and most probable valid support. The proposed MAP-List decoding is illustrated in Fig. 5.

VI. Numerical Results

In this section, we provide the simulation results that validate the error-correction capability of the proposed BOSS codes using the two-stage MAP decoding algorithm.

A. BLER Comparison

We evaluated different $(M = 128, B = 18)$ channel codes. Here, M and B denote the blocklength and the number of information bits mapped into each codeword, respectively. The resultant code rate is $\frac{18}{128} \approx 0.14$. Details on the code-design and decoding parameters of each code are provided below.

- **BOSS codes:** We simulated a two-layered BOSS code with blocklength $M = 128$; symmetric sparsity $K_1 = K_2 = 1$; candidate set cardinality $|\mathcal{M}^{(1)}| = 128$ and $|\mathcal{M}^{(2)}| = 64$; singleton PAM alphabets $\mathcal{A}_1 = \{1\}$ and $\mathcal{A}_2 = \{-1\}$; and $G = 32$. Hence, $\log_2(G) + \log_2(|\mathcal{M}^{(1)}|) + \log_2(|\mathcal{M}^{(1)}|) = 18$ bits are mapped into each BOSS codeword.
- **Polar codes:** We considered a polar code of length 128 and information set size $|\mathcal{A}_{\text{polar}}| = 18$, using the binary phase-shift keying (BPSK) modulation. The code construction is based on the 2×2 Arıkan kernel, and the 3GPP 5G NR polar sequence, i.e., pre-computed SNR-independent channel reliability order, [35] is adopted. SC and SCL decoding algorithm with search width $S = 16$ were employed. Here, the list size S is chosen to achieve the saturated ML-like performance, i.e., using S greater than 16 does not improve the SCL performance.
- **Tail-biting convolutional codes (TBCC):** A length-128 LTE legacy TBCC with BPSK modulation was evaluated under the wrap-around Viterbi algorithm (WAVA). The generator polynomials are $\mathbf{g} = \begin{bmatrix}133\ 171\ 165\end{bmatrix}$ in octal notation, with the mother code rate of $1/3$. In encoding, $3 \times 18 = 48$ coded bits are repeated to achieve the rate-matched output of length 128. The WAVA decoder first performs rate-recovery by soft-combining LLR values and then runs the Viterbi algorithm over the tail-biting trellis diagram.

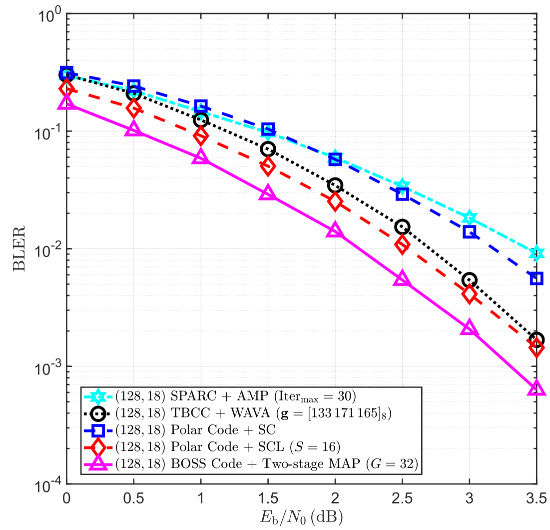

Fig. 6. BLER performance comparison with existing codes.

- **SPARCs:** We used a fat random Gaussian dictionary matrix $\mathbf{A}$ consisted of $U = 2$ sections with $\mathcal{C} = 512$ columns each to generate a SPARC of length 128, i.e., $\mathbf{A} \in \mathbb{R}^{128\times 1024}$. The SPARC was evaluated under AMP decoding with the maximum of 30 iterations, and we employed the online state evolution coefficient computation method proposed in [28]. While there exist multiple possible pairs of U and $\mathcal{C}$ in mapping 18 bits, we chose the combination that generates the largest dictionary matrix. Our line of reasoning is that AMP does not perform well with small-sized matrices, as its derivation from loopy belief propagation is based on the central limit theorem. As expected, we observed that the obtained SPARC yields the best performance among different combinations.

The results depicted in Fig. 6 demonstrate the superiority of BOSS codes over existing codes. The proposed two-stage MAP decoder delivers a substantial improvement, obtaining a coding gain of approximately 0.3 dB at a BLER of 10^{-2} compared to the polar code under SCL decoding.

B. Error-Correction Performance of CA-BOSS Codes

Fig. 7 compares the following codes of the same length 128 and rate $\frac{20}{128}$, including the state-of-the-art CA-polar and PAC codes that are known to very powerful in the short block-length regime. We have run extensive Monte-Carlo simulations to see when each code achieves the target BLER of 10^{-4}.

- **BOSS codes:** An $(M = 128, G = 128)$ BOSS code was tested under the two-stage MAP and MAP-List decoders with $S = 4$. Other code parameters are the same as in Fig. 6.
- **CA-BOSS codes:** The same BOSS code concatenated with the short CRC-3 outer code was simulated under MAP-List decoding with $S = 4$. The list was generated by selecting two most probable indices per layer.
- **CA-Polar codes:** Each input bit-stream of 20 bits is CRC-encoded and embedded into a carrier vector which is then fed into the polar encoder. We used the CRC-6

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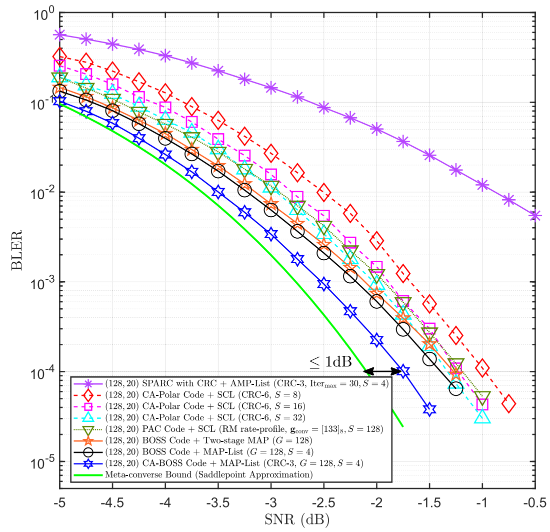

Fig. 7. BLER simulation results of $(128, 20)$ codes.

polynomial, so $|\mathcal{A}_{\text{polar}}| = 26$. The CA-SCL decoding algorithm was used with different list sizes $S \in \{8, 16, 32\}$.

- **PAC codes:** We constructed a PAC code by employing the Reed-Muller rate-profile and same generator polynomial $\mathbf{g}_{\text{conv}} = 133$ used in the original work of Arıkan [17]. SC-based list decoding [36] was used with the list size $S = 128$.
- **SPARCs with CRC:** A CRC-outer-coded SPARC with $U = 2$ and $\mathcal{C} = 1024$ was evaluated. To enhance the section error rate, which directly affects the BLER, CRC-encoding was performed on each section. We used the original AMP decoder followed by list decoding associated with the CRC code as proposed in [37]. The list size $S = 4$ was chosen to achieve the best performance.

Thought not significant, MAP-List decoding improves the original two-stage MAP decoder even without CRC, beating CA-polar under SCL with $S = 32$. By exploiting the CRC code, the performance of the MAP-List decoder is greatly enhanced: the CA-BOSS code outperforms both the CA-polar code and PAC code by a significant margin. If list decoding is employed without concatenating a CRC, the BOSS decoder may generate incorrect codeword candidates due to wrong support estimates in the first stage. However, the CRC test helps eliminate errors caused by these invalid estimates, reducing decoding complexity and improving performance through codeword pruning and error detection. Another observation worth to mention is that performance gains obtained by the increasing the list size in CA-polar codes gradually wane. This hints that the performance of CA-SCL with $S = 32$ is converging to the ML performance.

Fig. 7 also plots the saddlepoint approximation [38] to the meta-converse bound [15]:

$$\text{mc}(M, n) \triangleq \min_{P^M} \max_{Q^M} \{\alpha_{\frac{1}{n}}(P^M \times W^M, P^M \times Q^M)\}, \tag{39}$$

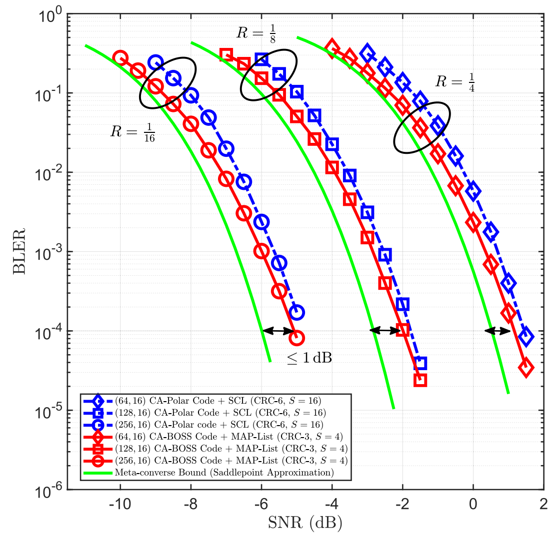

Fig. 8. Performance comparison between CA-BOSS and CA-polar codes when $R \in \{1/4, 1/8, 1/16\}$.

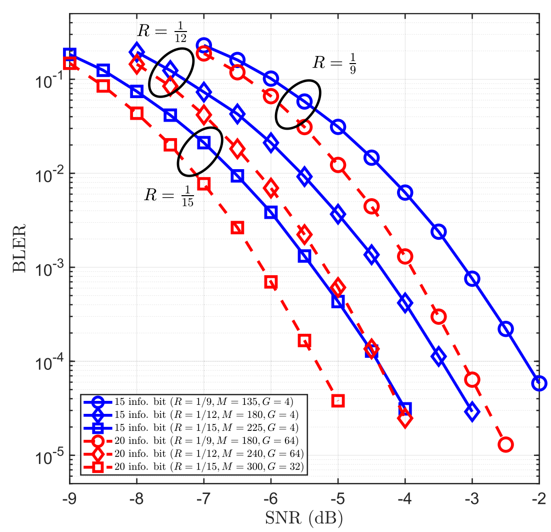

Fig. 9. BLER performances of CA-BOSS codes under MAP-List decoding with $S = 4$ when conveying $B \in \{15, 20\}$ bits for $R \in \{1/9, 1/12, 1/15\}$.

where n is the number of symbols transmitted over a length-M $W^M(\cdot|\cdot)$ channel, and $\alpha_\beta(P, Q)$ is the smallest tpye-I error probability across all tests between an input distribution P and auxiliary output distribution Q, with a type-II error probability of at most β. It can be seen that the CA-BOSS code performs within one dB away from the finite-length performance bound.

The comparison between CA-BOSS and CA-polar codes of the same information block size with different blocklengths $M \in \{64, 128, 256\}$ is depicted in Fig. 8. As can be seen, CA-BOSS codes have superior error-correction performance at all blocklengths, even when combined with a shorter CRC outer code. This demonstrates the exceptional error-correction capability of CA-BOSS codes. When the same information block size is transmitted, the addition of CRC bits results in the polar encoder using additional unreliable bit-channels, making the decoder more susceptible to errors during early stages of sequential decoding. However, this issue does not affect CA-BOSS coding. Moreover, the results indicate that CA-BOSS

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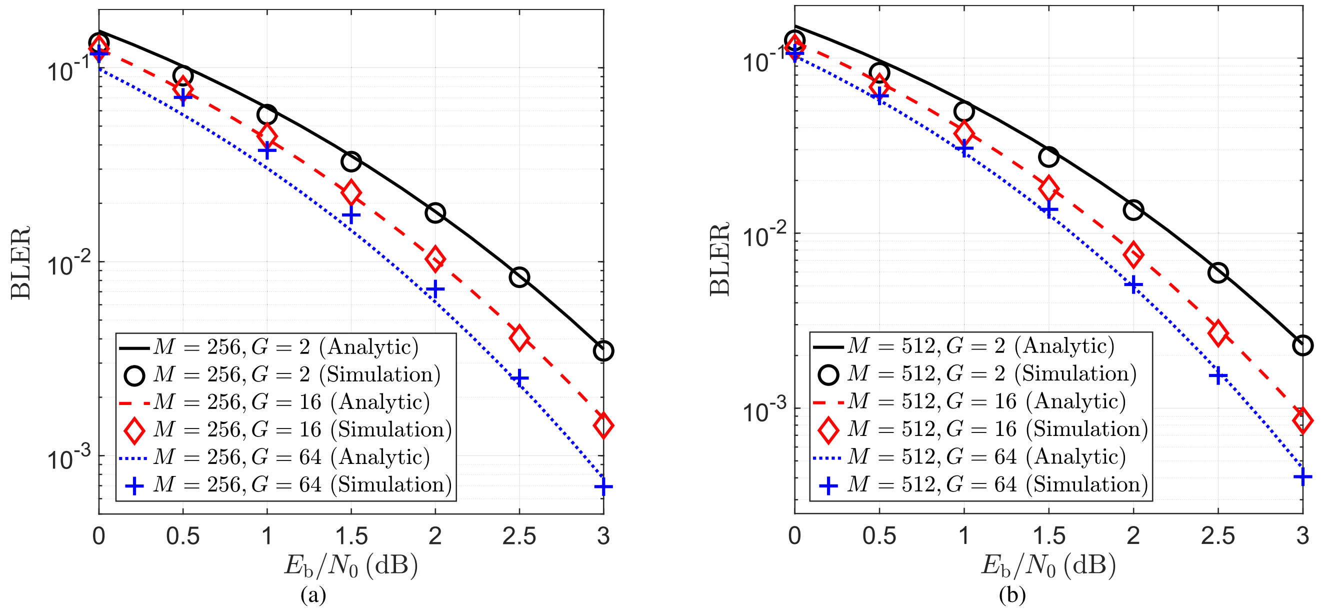

Fig. 10. BLER comparison between analytical and simulation results for $M \in \{256, 512\}$ and $G \in \{2, 16, 64\}$.

TABLE I
CODE PARAMETERS AND BIT-MAPPING RULES USED IN FIG. 9

B	R	CRC	M	$\lvert\mathcal{M}^{(1)}\rvert$	$\lvert\mathcal{M}^{(2)}\rvert$	G
15	$1/9$	$1+D+D^3$	135	128	64	4
	$1/12$		180	128	64	4
	$1/15$		225	128	64	4
20	$1/9$	$1+D+D^3$	180	128	128	64
	$1/12$		240	128	128	64
	$1/15$		300	256	128	32

codes perform close to the meta-converse bound in all SNRs and rates under consideration.

Fig. 9 illustrates the BLERs when sending fixed information bits, 15 or 20 bits, encoded using CA-BOSS codes with different low code rates, $\{1/9, 1/12, 1/15\}$. The simulation results demonstrate that CA-BOSS codes exhibit robust performance across a broad range of SNRs for various code rates. Hence, CA-BOSS codes can effectively support the transmission of various small information sizes, catering to a wide range of applications. For the purpose of replication, we have included the CRC polynomials and bit-mapping rules in Table I.

C. Validation of BLER Analysis

In Fig. 10, the comparison between the analytical BLER derived in Theorem 1 and the simulations is presented. The parameters used in the simulations are $M \in \{256, 512\}$, $\mathcal{A} = \{1\}$, and $G \in \{2, 16, 64\}$. The results indicate a close match between the analytical expression and the simulation results for different code parameters, validating the accuracy of our analysis. However, a small discrepancy can be observed in the low $E_{\rm b}/N_0$ regime. This discrepancy is due to numerical precision errors in computing $M(G-1)$ power in (37). As $M(G-1)$ increases, this discrepancy becomes more pronounced. However, it disappears as $E_{\rm b}/N_0$ increases.

VII. CONCLUSION

We have introduced a novel coded modulation method called BOSS codes for URLLC. The BOSS codes utilize a novel successive encoding technique to produce orthogonal codewords and a two-stage MAP decoding algorithm for low-latency decoding. The exact BLER expression for single-layered BOSS codes was provided to better understand the impact of code parameters on performance. The performance of BOSS codes was further improved through the concatenation with a CRC outer code, resulting in the CA-BOSS codes which achieved the meta-converse bound within 1 dB in the low code-rate regime.

Recent studies have demonstrated the robustness of BOSS codes in multi-path fading environments and showed their superiority over CA-polar codes with SCL decoding in real-time demonstrations using the IEEE 802.11 standards. Latency overhead induced by preambles may be non-negligible in URLLC applications. In this aspect, non-coherent detection methods for BOSS-coded SIMO and MIMO systems can be an important avenue for future research. Future work could also focus on expanding the capabilities of BOSS codes by incorporating phase shift keying modulation in complex AWGN channels for increased information delivery with energy efficiency.

APPENDIX
PROOF OF THEOREM 1

Before providing the proof, we first introduce two lemmas, which are necessary for our proof. The first lemma provides probability distribution of the maximum and minimum of M IID random variables.

Lemma 1: Let $\{X_m\}_{m=1}^{M}$ be a sequence of M IID random variables, each with probability density function (PDF) $f_X(x)$ and cumulative density function (CDF) $F_X(x)$. We also denote the maximum and minimum

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of the sequence by $X_{\max} = \max_{1\le m\le M} X_m$ *and* $X_{\min} = \min_{1\le m\le M} X_m$*, respectively. Then, the PDF's of* $X_{\max}$ *and* $X_{\min}$ *are*

$$f_{X_{\max}}(x) = Mf_X(x)F_X(x)^{M-1} \quad \text{and} \tag{40}$$

$$f_{X_{\min}}(x) = Mf_X(x)[1-F_X(x)]^{M-1}. \tag{41}$$

Proof: Due to independence, the CDF of $X_{\max}$ can be readily found:

$$F_{X_{\max}}(x) = \prod_{m=1}^{M} \mathbb{P}(X_m \le x) = F_X(x)^M. \tag{42}$$

By differentiating $F_{X_{\max}}(x)$ with respect to x, we obtain $f_{X_{\max}}(x)$. Similarly, $F_{X_{\min}(x)} = 1-[1-F_X(x)]^M$, and the rest of the proof is straightforward.

The second lemma provides statistics of an inner product of two IID random unit vectors in an $(M-1)$-dimensional sphere.

Lemma 2: For two random unit-norm vectors, $\mathbf{u} \in \mathbb{R}^M$ and $\mathbf{v} \in \mathbb{R}^M$, that are uniformly distributed on a sphere S^{M-1}, their inner product $W = \langle \mathbf{u}, \mathbf{v}\rangle$ has a PDF given by

$$f_W(w) = \frac{\Gamma\left(\frac{M}{2}\right)}{\sqrt{\pi}\,\Gamma\left(\frac{M-1}{2}\right)}(1-w^2)^{\frac{M-3}{2}}. \tag{43}$$

Proof: If $\mathbf{X} = \{X_1, X_2, \ldots X_M\}$ follows the standard multivariate normal distribution, i.e., $X_i \sim \mathcal{N}(0,1)$, then $\frac{\mathbf{X}}{\|\mathbf{X}\|_2}$ is uniformly distributed on the unit sphere. We assume that $\mathbf{v} = [1, 0, \ldots, 0]$ thanks to spherical symmetry, and then the distribution of W is identical to that of

$$\frac{X_1}{\sqrt{X_1^2 + X_2^2 + \cdots + X_M^2}}. \tag{44}$$

It is well known that the square of ratio distribution in (44) follows Beta$(\frac{1}{2}, \frac{M-1}{2})$. The rest of the proof is trivial.

Now, we are ready to prove Theorem 1. We denote by $\mathcal{E}_1$ and $\mathcal{E}_2$ an error event at stage 1 and 2 of the proposed two-stage MAP decoder, respectively. Using these error events, the probability of decoding error is given by

$$\mathbb{P}(\mathcal{E}) = \mathbb{P}(\mathcal{E}_1) + \mathbb{P}(\mathcal{E}_2|\mathcal{E}_1^c)\mathbb{P}(\mathcal{E}_1^c). \tag{45}$$

We consider a randomly constructed dictionary matrix $\mathbf{A} = [\mathbf{U}_1, \mathbf{U}_2, \ldots, \mathbf{U}_G]$, i.e., $\mathbf{U}_i$ and $\mathbf{U}_j$ are independent for $i \neq j \in [G]$. Without loss of generality, we assume that $K_1 = 1$ and $\mathbf{U}_1$ has participated in encoding, so $\mathbf{A}\mathbf{x} = \mathbf{U}_1\mathbf{x}_1$, where $\|\mathbf{x}_1\|_0 = 1$. For the AWGN channel, the received symbol in the mth channel use is given by

$$Y_m = \begin{cases} 1 + V_m & \text{for } m \in \mathcal{I} \\ V_m & \text{for } m \notin \mathcal{I}, \end{cases} \tag{46}$$

where V_m is a zero-mean Gaussian noise with variance σ_v^2. As explained in Remark 4, a simple OS decoder is optimal in this case. Then, the error event at stage 1 is $\{\hat{\mathbf{x}}_1 \neq \mathbf{x}_1\}$, and its probability can be computed as

$$\begin{aligned}\mathbb{P}(\mathcal{E}_1) &= 1 - \mathbb{P}(\mathcal{E}_1^c) \\ &= 1 - \mathbb{P}(Y_{\mathcal{I}}^{\min} > Y_{\mathcal{I}^c}^{\max}),\end{aligned} \tag{47}$$

where $Y_{\mathcal{I}}^{\min} \triangleq \min_{m\in\mathcal{I}} Y_m$ and $Y_{\mathcal{I}^c}^{\max} \triangleq \max_{m\in\mathcal{I}^c} Y_m$. The conditional expectation theorem gives

$$\begin{aligned}&\mathbb{P}(Y_{\mathcal{I}}^{\min} > Y_{\mathcal{I}^c}^{\max}) \\ &= \mathbb{E}_{Y_{\mathcal{I}^c}^{\max}}\left\{\mathbb{P}(Y_{\mathcal{I}}^{\min} > y)|Y_{\mathcal{I}^c}^{\max} = y\right\} \\ &= \int_{-\infty}^{\infty} \mathbb{P}(Y_{\mathcal{I}}^{\min} > y) f_{Y_{\mathcal{I}^c}^{\max}}(y)\mathrm{d}y \\ &= \int_{-\infty}^{\infty} \mathbb{P}(Y_{\mathcal{I}}^{\min} > y)(M-1)f_{Y_{\mathcal{I}^c}}(y)F_{Y_{\mathcal{I}^c}}(y)^{M-2}\mathrm{d}y \\ &= \frac{(M-1)}{\sqrt{2\pi\sigma_v^2}}\int_{-\infty}^{\infty}[1-F_{Y_{\mathcal{I}}}(y)]F_{Y_{\mathcal{I}^c}}(y)^{M-2}e^{-\frac{y^2}{2\sigma_v^2}}\mathrm{d}y.\end{aligned} \tag{48}$$

Note that Y_m is distributed as $\mathcal{N}(1, \sigma_v^2)$ for $m \in \mathcal{I}$ and $\mathcal{N}(0, \sigma_v^2)$ for $m \in \mathcal{I}^c$. Plugging $F_{Y_{\mathcal{I}}}(y) = 1 - Q\left(\frac{y-1}{\sigma_v}\right)$ and $F_{Y_{\mathcal{I}^c}}(y) = 1 - Q\left(\frac{y}{\sigma_v}\right)$ into (48), we arrive at the expression in (36).

Now, we shift our focus to the second stage. Under the condition that the decoder has correctly estimated a sparse sub-message vector under $\mathcal{H}_1$, i.e., $\hat{\mathbf{x}}_1 = \mathbf{x}_1$, an error occurs when at least one codeword estimate $\mathbf{U}_g\hat{\mathbf{x}}_g$ for $g \in [G]\backslash\{1\}$ is closer to $\mathbf{y}$ in terms of Euclidean distance compared to $\mathbf{U}_1\hat{\mathbf{x}}_1$. The conditional error probability at stage 2 can be expressed as follows:

$$\begin{aligned}&\mathbb{P}(\mathcal{E}_2|\mathcal{E}_1^c) \\ &\quad= 1 - \mathbb{P}(\hat{g} = 1|\hat{\mathbf{x}}_1 = \mathbf{x}_1) \\ &\quad= 1 - \mathbb{P}\left(\cap_{g\in[G]\backslash\{1\}}\{\|\mathbf{y} - \mathbf{U}_1\hat{\mathbf{x}}_1\|_2 \right. \\ &\qquad\qquad\qquad \left. < \|\mathbf{y} - \mathbf{U}_g\hat{\mathbf{x}}_g\|_2\}|\hat{\mathbf{x}}_1 = \mathbf{x}_1\right) \\ &\quad= 1 - \mathbb{P}\left(\cap_{g\in[G]\backslash\{1\}}\{\mathbf{y}^{\mathsf{T}}\mathbf{U}_1\mathbf{x}_1 > \mathbf{y}^{\mathsf{T}}\mathbf{U}_g\hat{\mathbf{x}}_g\}\right).\end{aligned} \tag{49}$$

We first consider the probability that $\mathbf{y}^{\mathsf{T}}\mathbf{U}_1\mathbf{x}_1$ is larger than $\mathbf{y}^{\mathsf{T}}\mathbf{U}_g\hat{\mathbf{x}}_g$. Recall that the OS decoder estimates the support under $\mathcal{H}_g$ by finding the maximum element of $\mathbf{U}_g^{\mathsf{T}}\mathbf{y} = \mathbf{U}_g^{\mathsf{T}}(\mathbf{U}_1\mathbf{x}_1 + \mathbf{v})$. We define the mth element of random vector $\mathbf{U}_g^{\mathsf{T}}\mathbf{y}$ as

$$\tilde{Y}_{g,m} = \mathbf{u}_{g,m}^{\mathsf{T}}\mathbf{v} + \mathbf{u}_{g,m}^{\mathsf{T}}\mathbf{U}_1\mathbf{x}_1. \tag{50}$$

The first term $\tilde{V}_{g,m} = \mathbf{u}_{g,m}^{\mathsf{T}}\mathbf{v}$ is a zero-mean Gaussian random variable with variance σ_v^2, while the second term $W_{g,m} = \mathbf{u}_{g,m}^{\mathsf{T}}\mathbf{U}_1\mathbf{x}_1$ is distributed per Lemma 2. The decoder finds the maximum index:

$$\hat{\mathcal{I}}_g = \arg\max_{m\in[M]} \mathbf{u}_{g,m}^{\mathsf{T}}\mathbf{U}_1\mathbf{x}_1 + \mathbf{u}_{g,m}^{\mathsf{T}}\mathbf{v}. \tag{51}$$

The decoder generates $\hat{\mathbf{x}}_g$ as a one-hot vector with a non-zero value at $\hat{\mathcal{I}}_g$. Therefore, $\mathbf{y}^{\mathsf{T}}\mathbf{U}_g\hat{\mathbf{x}}_g$ is distributed as the maximum of $\tilde{Y}_{g,m}$, i.e., $\max\left\{\tilde{Y}_{g,1}, \ldots, \tilde{Y}_{g,M}\right\}$, and (49) can be re-

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written as

$$\begin{aligned}
&\mathbb{P}(\mathcal{E}_2|\mathcal{E}_1^c)\\
&= 1-\mathbb{P}\left(\bigcap_{g\in[G]\backslash\{1\}}\left\{\mathbf{y}^{\mathsf{T}}\mathbf{U}_1\mathbf{x}_1 > \max\left\{\tilde{Y}_{g,1},\ldots,\tilde{Y}_{g,M}\right\}\right\}\right)\\
&= 1-\mathbb{P}\left(\bigcap_{\substack{g\in[G]\backslash\{1\}\\ m\in[M]}}\left\{\mathbf{y}^{\mathsf{T}}\mathbf{U}_1\mathbf{x}_1 > \tilde{Y}_{g,m}\right\}\right)\\
&= 1-\mathbb{P}\left(\bigcap_{\substack{g\in[G]\backslash\{1\}\\ m\in[M]}}\left\{\mathbf{v}^{\mathsf{T}}\mathbf{U}_1\mathbf{x}_1 + \|\mathbf{U}_1\mathbf{x}_1\|_2^2 > \tilde{Y}_{g,m}\right\}\right).
\end{aligned} \tag{52}$$

Note that $\tilde{Y}_{g,m} = \tilde{V}_{g,m} + W_{g,m}$ is IID for $g \in [G]\backslash\{1\}$. Denoting $\tilde{Z}_{g,m} = \mathbf{v}^{\mathsf{T}}\mathbf{U}_1\mathbf{x}_1 - \tilde{V}_{g,m}$, a zero-mean Gaussian with variance $2\sigma_v^2$, we obtain the following expression:

$$\begin{aligned}
&\mathbb{P}(\mathcal{E}_2|\mathcal{E}_1^c)\\
&\quad = 1-\prod_{\substack{g\in[G]\backslash\{1\}\\ m\in[M]}}\mathbb{P}(\tilde{Z}_{g,m} > W_{g,m}-1)\\
&\quad = 1-\prod_{\substack{g\in[G]\backslash\{1\}\\ m\in[M]}}\mathbb{E}_{W_{g,m}}\left\{\mathbb{P}(\tilde{Z}_{g,m} > w-1)|W_{g,m}=w\right\}\\
&\quad = 1-\left[\frac{\Gamma\left(\frac{M}{2}\right)}{\sqrt{\pi}\Gamma\left(\frac{M-1}{2}\right)}\right.\\
&\qquad\quad \left.\times\int_{-1}^{1}Q\left(\frac{w-1}{\sqrt{2\sigma_v^2}}\right)(1-w^2)^{\frac{M-3}{2}}\mathrm{d}w\right]^{M(G-1)}.
\end{aligned} \tag{53}$$

Finally, by plugging (36) and (53) into (45), we complete the proof.

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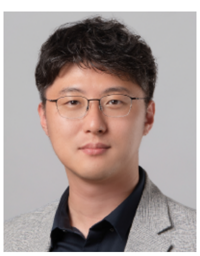

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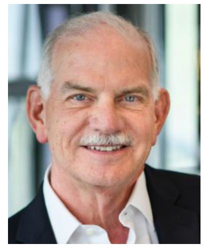

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